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Mass Movement Hazard and Exposure in the Himalaya

Quantifying how avalanches, rockfalls, and landslides threaten buildings, roads, waterways, population, and glacial lakes across the Hindu Kush Karakoram Himalaya (HKKH) today and in 2100.

AUTHOR

Dr. Saket Dubey, Dr. Ashim Sattar, Dr. Umesh K. Haritashya, and Dr. Christian Huggel, Prof. Manish Kumar Goyal, Dr. Simon Allen, Dr. Holger Frey

Prof. Manish Kumar Goyal
Professor of Civil Engineering
Indian Institute of Technology, Indore

mkgoyal@iiti.ac.in +91 9435880989

Karakoram Has the Densest Mass Movement Hazard

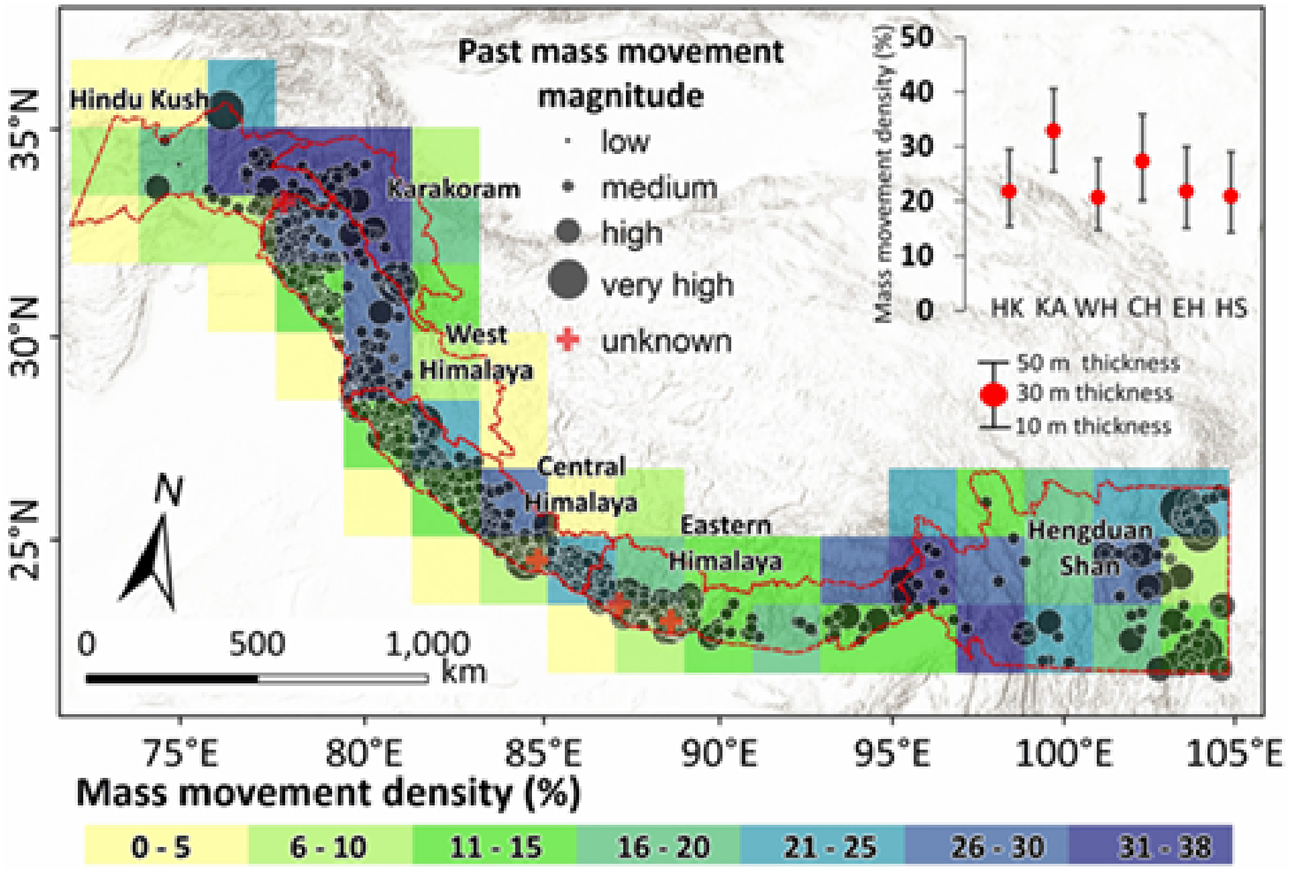


Figure 2. Modeled mass movement density (%) across the HKKH (30 m failure depth), with past landslide events overlaid.

fatal landslide events recorded across the HKKH between 2007 and 2016

1,339

of those events struck Central Himalaya (41%) and Western Himalaya (27%) alone

41% + 27%

has the highest mass movement density of any range, with Central Himalaya close behind — a clear match with the historical event record

Karakoram

Karakoram's Roads vs. Central Himalaya's Rivers

The team screened over 3.6 million buildings, 0.2 million km of waterways, and 0.5 million km of road network across the HKKH for intersection with modeled mass movement paths.

Karakoram (KA)

has the highest percentage of exposed buildings and roadways — despite having only 3.3% of Central Himalaya's building density and the lowest road density of any range

Central Himalaya (CH)

has the highest exposed waterways (0.37 km/km² density) — the most river-dense range in the HKKH

Exposed roadway hotspots:

Nepal · Uttarakhand · Himachal Pradesh · Ladakh · China's Sichuan Province

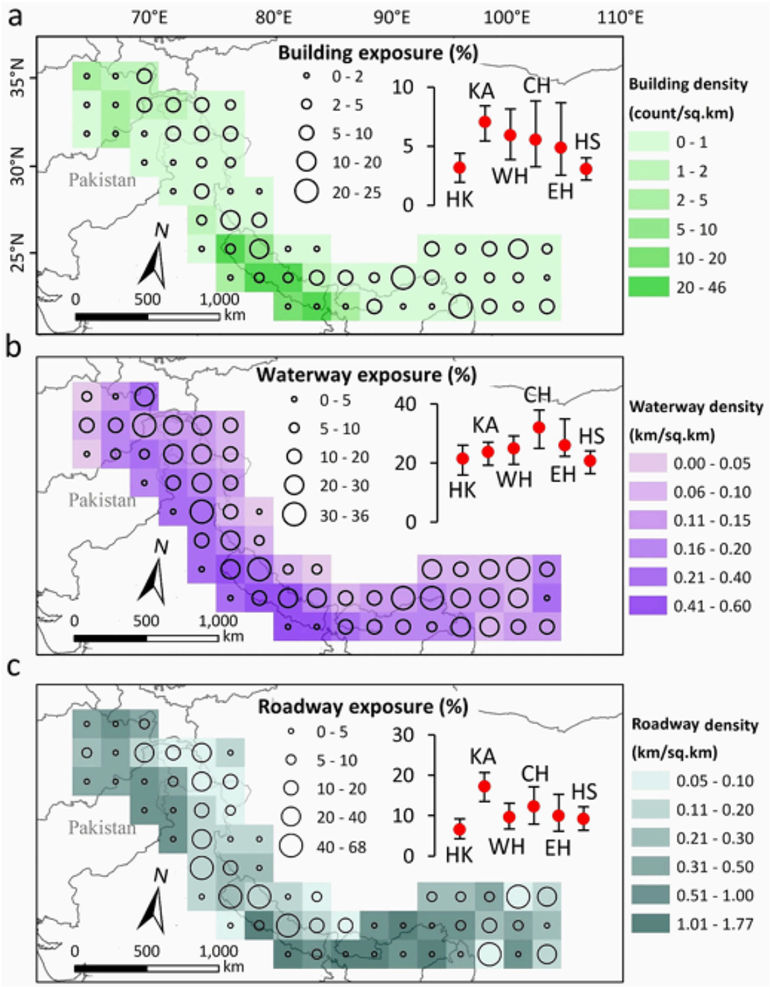
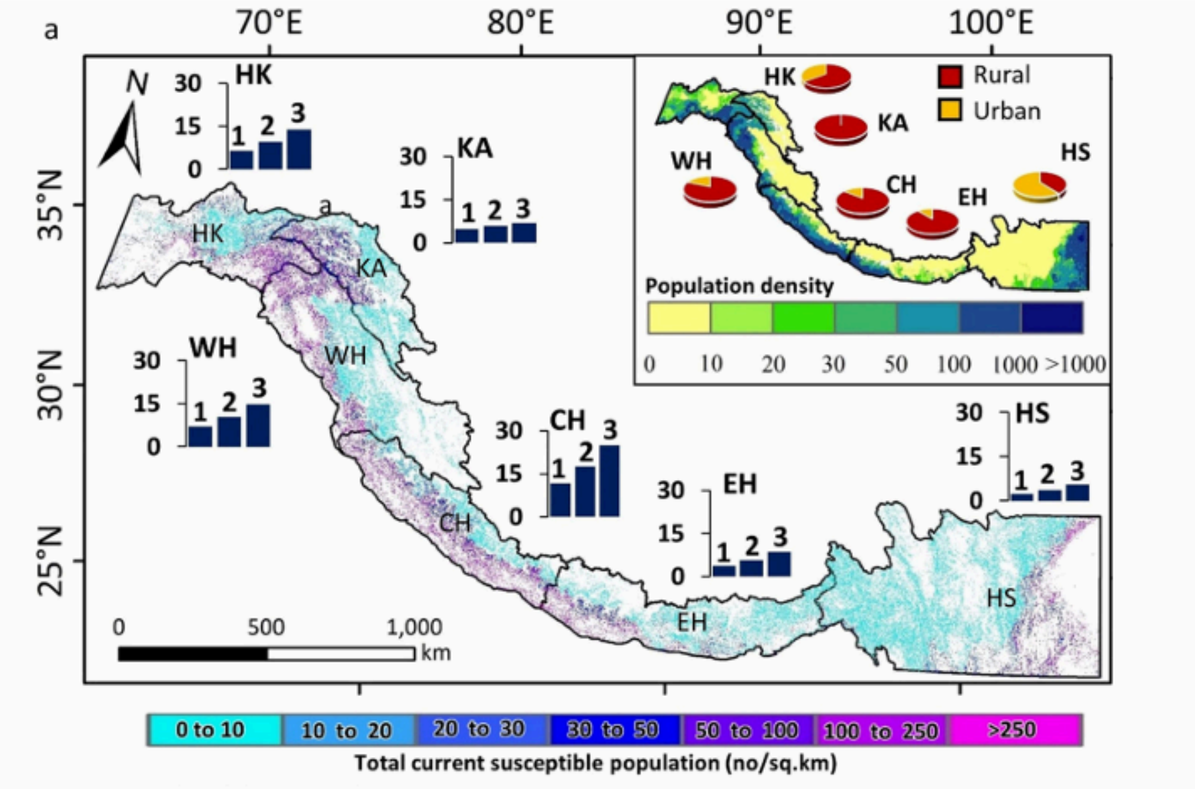


Figure 3. Exposure (%) of buildings, waterways, and roadways to mass movements, 30 m failure depth.

110.8 Million People Live in the Hazard Zone



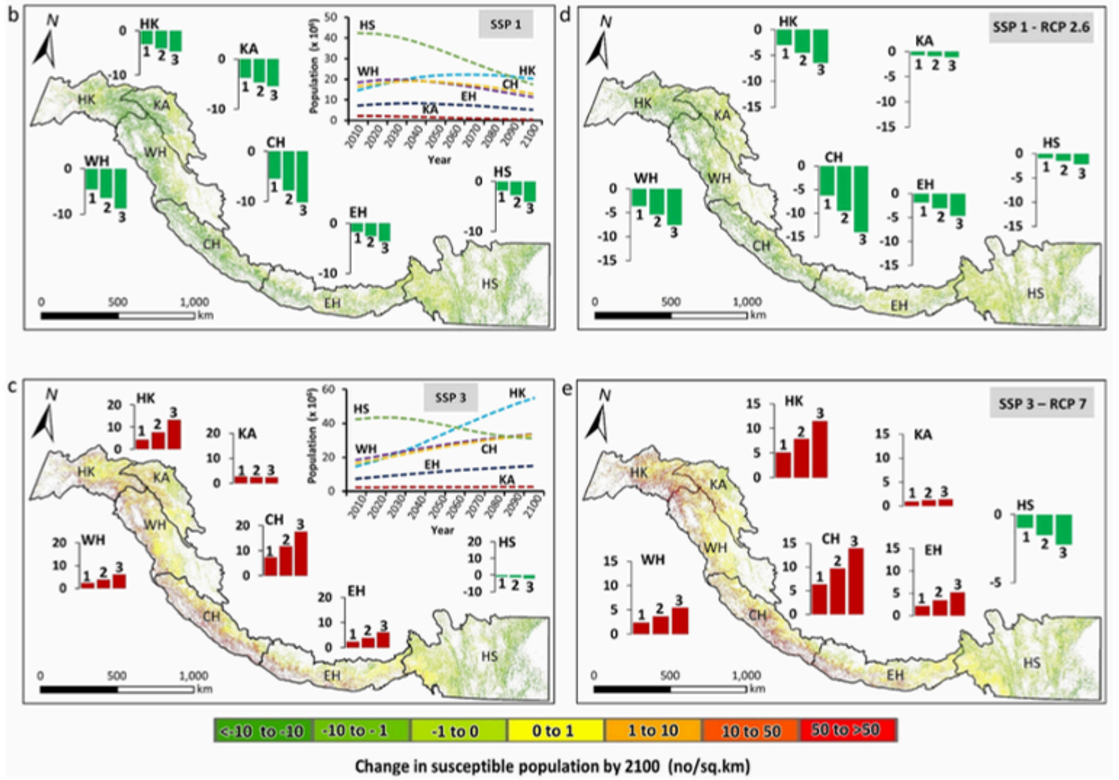


Figure 5a. Current population (people/km²) exposed to mass movements, with rural/urban split by range (30 m failure depth).

average exposed population as failure depth rises from 10 m to 50 m

6 → 13 / km²

has the highest current population exposure of all six mountain ranges

Central Himalaya

gap between the lowest (SSP1) and highest (SSP3) 2100 population scenarios

103.3 million

Retreating Glaciers Are Creating New Flood Risks

When a mass movement strikes a glacial lake, it can trigger a displacement wave that overtops or breaches the dam — a GLOF. The study traced which lakes, today and in the future, sit in the path.

of 3,601 existing glacial lakes (>0.05 km²) are susceptible to mass-movement impact today

70%

has the highest current exposure and the most headward (direct, high-impact) lake strikes

Eastern Himalaya

more than any other range — shifting the future GLOF hotspot from Eastern Himalaya to Karakoram

832 new lakes in Karakoram

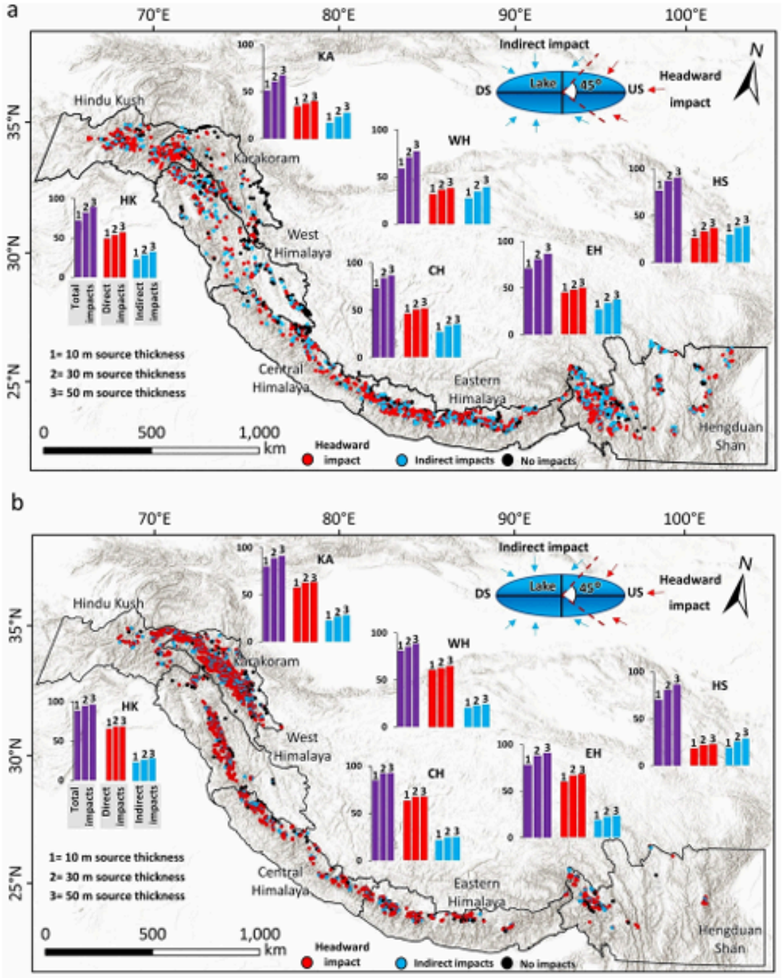


Figure 6. Mass movement susceptibility of (a) existing and (b) future glacial lakes, 30 m failure depth.